

Aleksandr Turchaninov

AI ENGINEER · BACKEND ENGINEER (GO)

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PORTFOLIO

Work authorisation: Stamp 2 (20 h/week in term, full-time in holidays) · MSc completes Feb 2027, then eligible for Stamp 1G (Third Level Graduate Programme)

SUMMARY

Backend engineer (Go) with 3.5 years of commercial experience, working where backend meets LLM and ML infrastructure. I design, build and run RAG systems, LLM pipelines and high-load microservices end to end — taking production services from architecture to release independently and aligning decisions with the tech lead and stakeholders. Currently completing an MSc in Computing at ATU Galway. English C1.

EDUCATION

Atlantic Technological University, Galway — MSc in Computing expected Feb 2027

Thesis: satellite-based detection of industrial turf-cutting in legally protected Irish peatlands. Modules include Machine Learning, Statistical Computing, Text & Sequence Analysis, Cloud Native Computing, Advanced Databases, Research Methods.

Bauman Moscow State Technical University — BSc in Software Engineering Sep 2021 – Jun 2025

EXPERIENCE

Ozon Tech — AI Engineer / Backend Engineer (Go), remote Jul 2025 – Present

Team of 6 (3 engineers). ML-integration services and RAG pipelines for internal products used by ~5,000 employees.

HR-bot — RAG assistant over the internal knowledge base · designed and owned from scratch

- Delivered a RAG search system over Confluence and internal docs to production single-handedly: ~100 queries/day, ~95% correct answers, ~90% user satisfaction.
- Hybrid search (Qwen embeddings + full-text) on PostgreSQL / pgvector; LLM orchestration on Eino (Go), migrated from an early langchaingo version for graph/workflow patterns.
- LLM tracing via Langfuse, Mattermost integration, automatic knowledge-base sync from Confluence. Extended into SD-bot, a multi-tenant layer where tenant-specific logic lives in config.
- Go, gRPC + grpc-gateway, pgx / sqlc, Kafka, Eino, pgvector.

HR-document-ML — gateway to ML document recognition · owned

- Proxy service recognising ~25 document types (passports, travel documents, certificates), with an S3 proxy fronting storage behind HTTP handlers.
- Designed the personal-data flow so the service never stores or sees raw PII: data passes to secure storage in transit and only contracted fields are returned.
- Built a fine-tuning loop where HR corrections become the model's training set.

Meeting Scribe and Thread Summary — transcription and summarisation

- Asynchronous pipeline on worker pods: PostgreSQL task queue, leader election, chunking of 30-minute to 2-hour meetings, heartbeat-based recovery of stuck tasks.
- WhisperX transcription with speaker diarisation, FFmpeg audio processing (multi-track Opus / .mka), voice-track encryption, Prometheus metrics.
- Thread summarisation for ~500 threads/day, including incident threads of 2,000+ messages via split-and-merge.

Internal platforms: the WINK streaming service, then a video-analytics platform.

- Video analytics: built video-stream synchronisation with asynchronous forced sync; migrated Go services from RabbitMQ to Kafka with end-to-end pipeline testing; introduced TLS across platform services.
- Streaming: optimised the frontend–backend API, added caching for high-load services, JWT authentication and authorisation, SQL query optimisation, monitoring and logging.
- Added SBOM generation to GitLab CI; test coverage, code review and task grooming.

SELECTED SIDE PROJECTS

- **REPRESENT CRM** — AI sales assistant and CRM for a clothing store on Avito: Claude answers customer chats, the CRM tracks leads and listings.
- **Lily** — private app for couples on iOS (SwiftUI) and Android: shared places, films, trips and moments.
- **Vestochka** — Telegram Mini App that turns a photo into a watercolour greeting card you can spin in 3D.

SKILLS

Languages	Go, Python, SQL
LLM / AI	RAG, pgvector, hybrid search, embeddings, Eino, langchaingo, LLM orchestration, WhisperX, Langfuse
Data / brokers	PostgreSQL, Redis, Kafka, RabbitMQ
API / infra	gRPC, grpc-gateway, REST, Docker, Kubernetes, Linux, Git, GitLab CI
Tools	pgx, sqlc, fx (DI), Prometheus, etcd, FFmpeg
Human languages	Russian (native), English (C1)